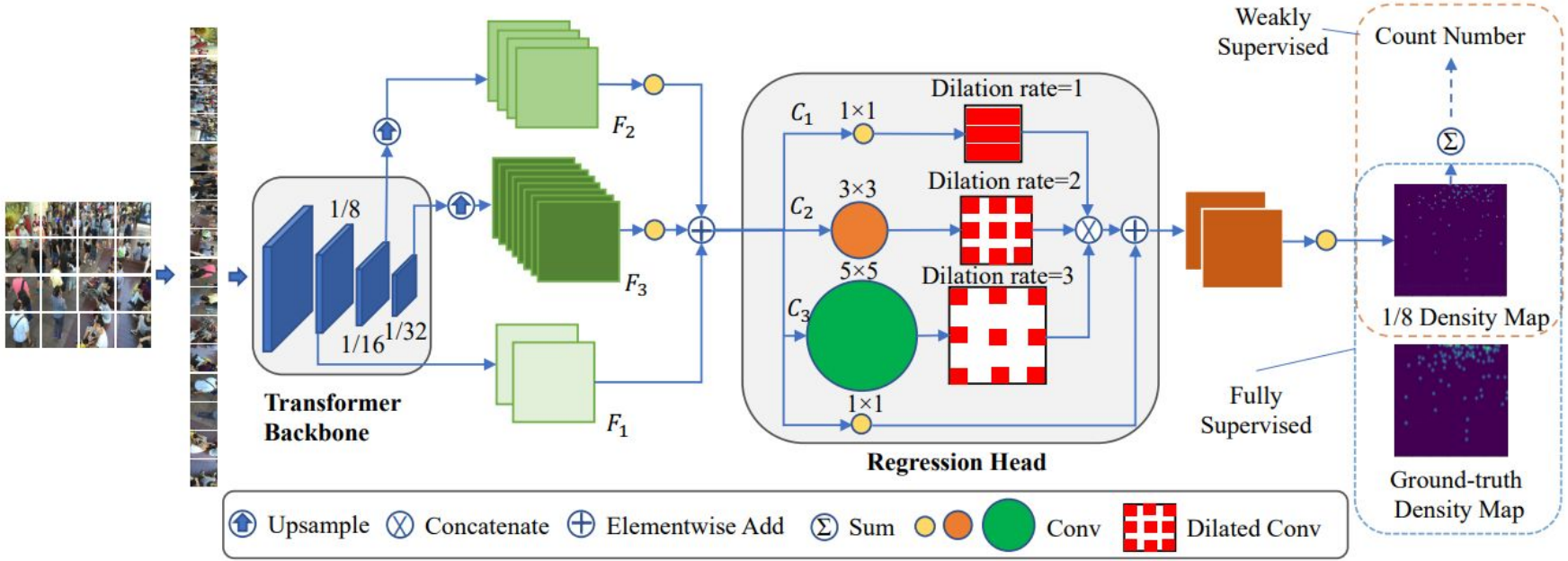
CCTrans: Crowd Counting with Transformer

Authors: Ye Tian, Xiangxiang Chu, Hongpeng Wang

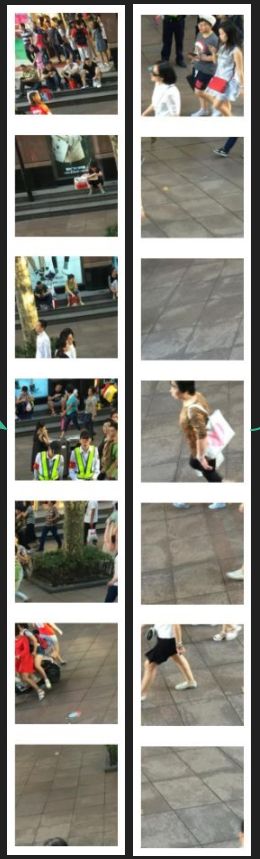
Overview


Weakly Supervised
Count Number
Dilation rate=1
C_1
1×1
F_2
Dilation rate=2
C_2
3×3
1/8
F_3
5×5
Dilation rate=3
C_3
1/16
1/32
1/8 Density Map
Transformer Backbone
F_1
1×1
Fully Supervised
Regression Head
Ground-truth Density Map
Upsample
Concatenate
Elementwise Add
Sum
Conv
Dilated Conv

2D image to 1D sequence

Size: H*W*3

Each of size K*K*3



Patches flattened into vectors of size D = K*K*3

$x_1, x_2, x_3 \ldots\ldots x_N.$

where $N = (H*W)/K^2$ and $x_i \in R^D$

$f(x): x_i \rightarrow e_i$

$e = e_1, e_2, e_3 \ldots\ldots e_N.$

where $e_i \in R^D$ and $e \in R^{D*N}$

Patch embedding to derive meaningful features

f(x) : learnable projection

In this way, spatial and channel features of the ith image patch x_i are transformed into embedded features of the ith embedding vector e_i .

Transformer Backbone

Uses Pyramid Transformer Backbone: Twins-SVT (Xiangxiang Chu, Zhi Tian, Yuqing Wang , Bo Zhang, Haibing Ren, Xiaolin Wei1 , Huaxia Xia , Chunhua Shen)

Adds 2 main concepts:

LSA: locally-grouped self-attention

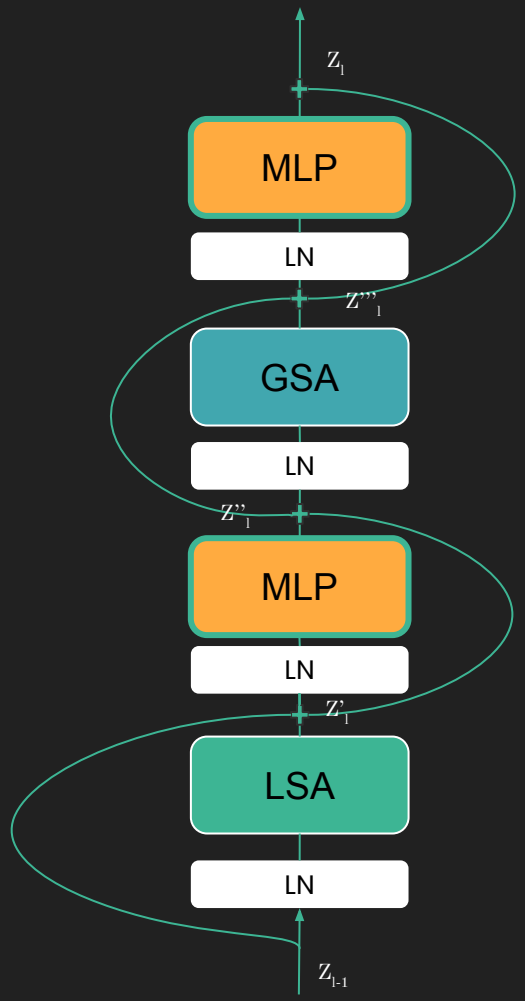
GSA: Global sub-sampled attention

$Z'_l = LSA(LN(Z_{l-1})) + Z_{l-1}$ (1)

$Z''_l = MLP(LN(Z'_l)) + Z'_l$ (2)

$Z'''_l = GSA(LN(Z''_l)) + Z_l$ (3)

$Z_l = MLP(LN(Z'''_l)) + Z_l$. (4)



Layer L

Self-attention - 1

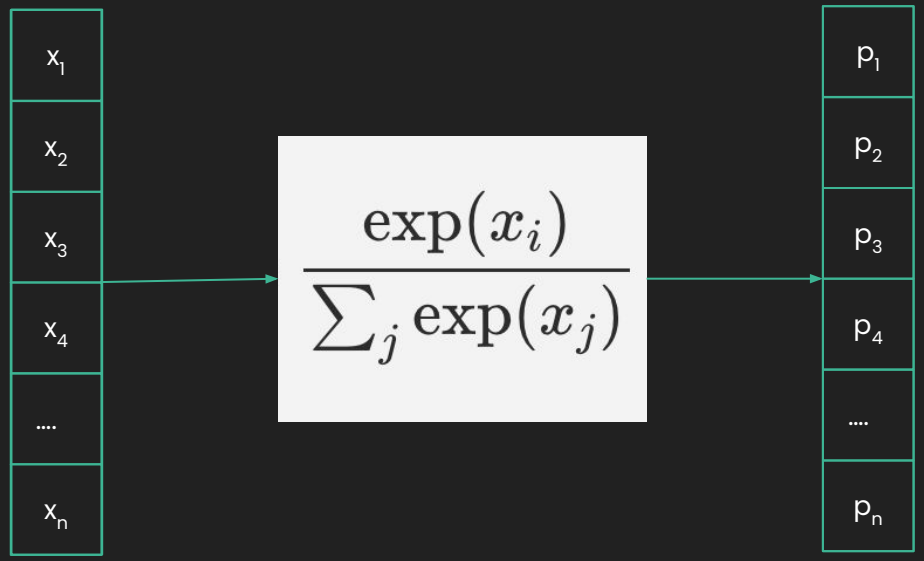
Self-attention allows model to relate the patches with each other

$$\text{Attention}(Q,K,V) = \text{softmax}((Q.K^T)/\text{sqrt}(D)).V$$

$$\text{softmax}\left(\frac{Q_{NxD} \times K^T_{DxN}}{\text{sqrt}(D)}\right) =$$

	e_1	e_2	e_3	....	e_n
e_1	$e_1.e_1$	$e_1.e_2$	$e_1.e_3$	....	$e_1.e_n$
e_2	$e_2.e_1$	$e_2.e_2$	$e_2.e_3$	....	$e_2.e_n$
e_3	$e_3.e_1$	$e_3.e_2$	$e_3.e_3$	....	$e_3.e_n$
....	....	....	....	....	....
e_n	$e_n.e_1$	$e_n.e_2$	$e_n.e_3$	....	$e_n.e_n$

Softmax function



Convert raw attention scores to probabilities

Emphasize important tokens and suppress irrelevant ones

Self-attention - 2

	e_1	e_2	e_3	....	e_n
e_1	$e_1.e_1$	$e_1.e_2$	$e_1.e_3$	....	$e_1.e_n$
e_2	$e_2.e_1$	$e_2.e_2$	$e_2.e_3$	....	$e_2.e_n$
e_3	$e_3.e_1$	$e_3.e_2$	$e_3.e_3$	....	$e_3.e_n$
....	....	....	....	....	....
e_n	$e_n.e_1$	$e_n.e_2$	$e_n.e_3$	....	$e_n.e_n$

$\times$

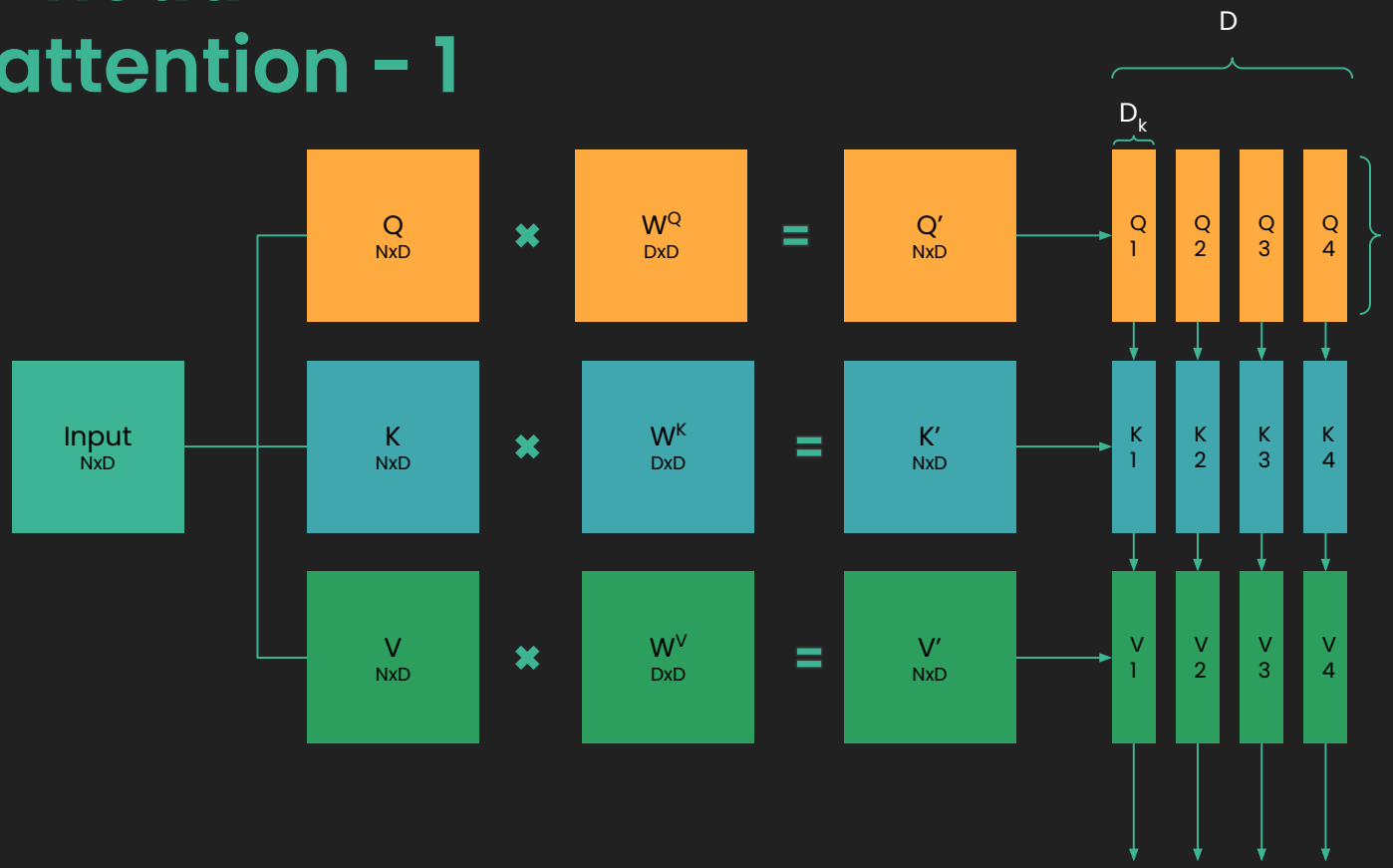
V
NxD

=

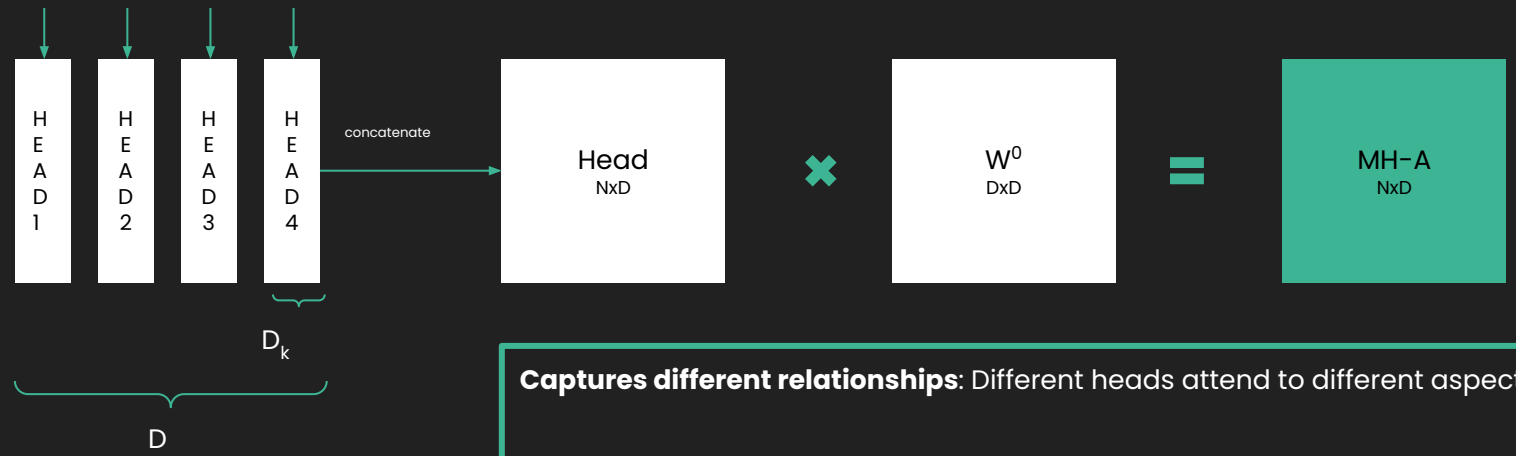
Attention
NxD

Each row in the attention matrix captures not only the meaning of each patch but also relationship of each patch with other patches

Multi-head Self-attention - 1



Multi-head Self-attention - 2



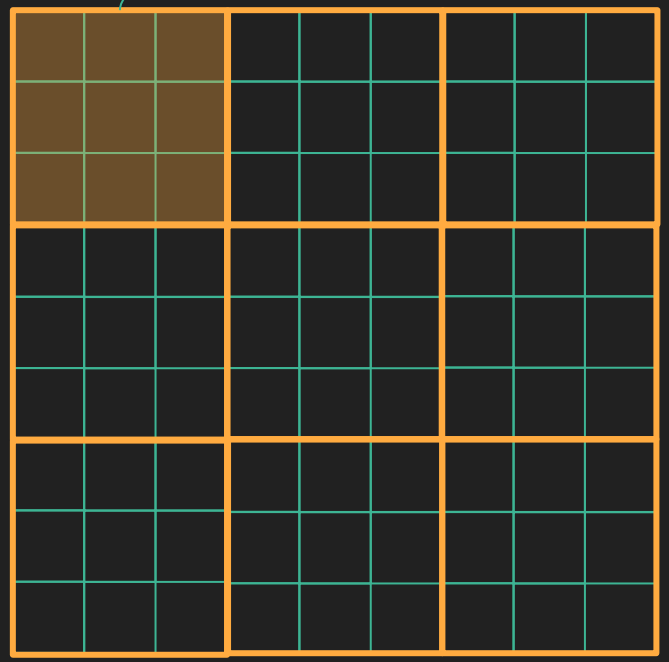
$$\mathrm{Head}_i = \mathrm{Attention}(Q_i, K_i, V_i)$$

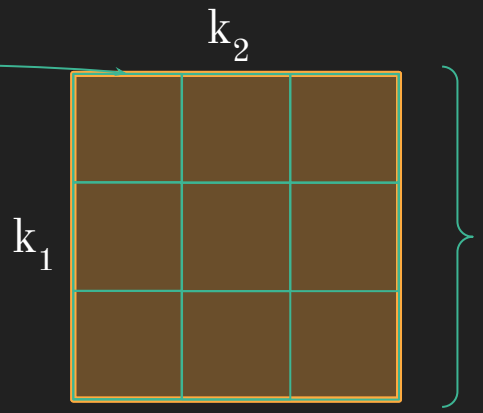
Captures different relationships: Different heads attend to different aspects of the input.

Improves learning efficiency: By operating in parallel, multiple heads allow for better learning of dependencies.

Enhances robustness: The model doesn't rely on a single attention pattern, reducing overfitting.

Locally-grouped self-attention



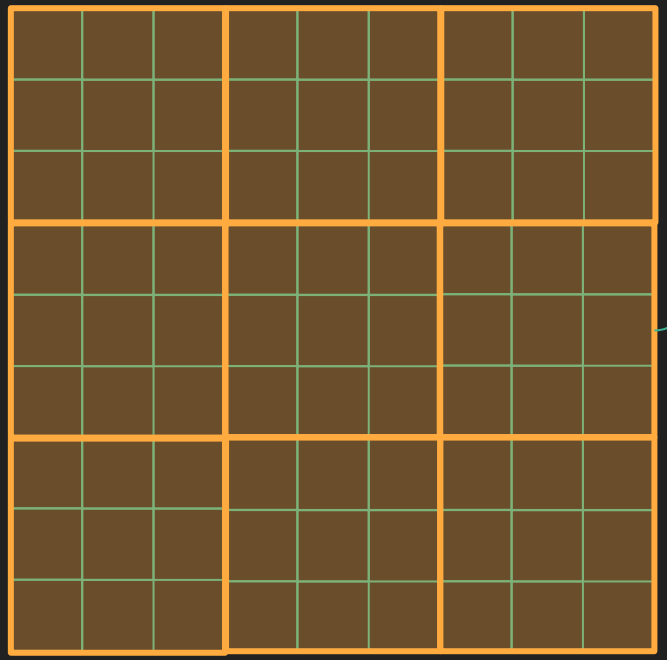


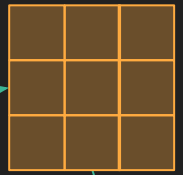
Sub-windows of size: $k_1 \times k_2$ are processed.

Followed by self-attention of these sub-windows.

Helpful in attain local features computationally faster.

Global sub-sampled attention





Used as key in self-attention. Then upscaled

sub-window

1	2	3
4	5	6
7	8	9

*

kernel

1	0	-1
1	0	-1
1	0	-1

Convolution Example

-6

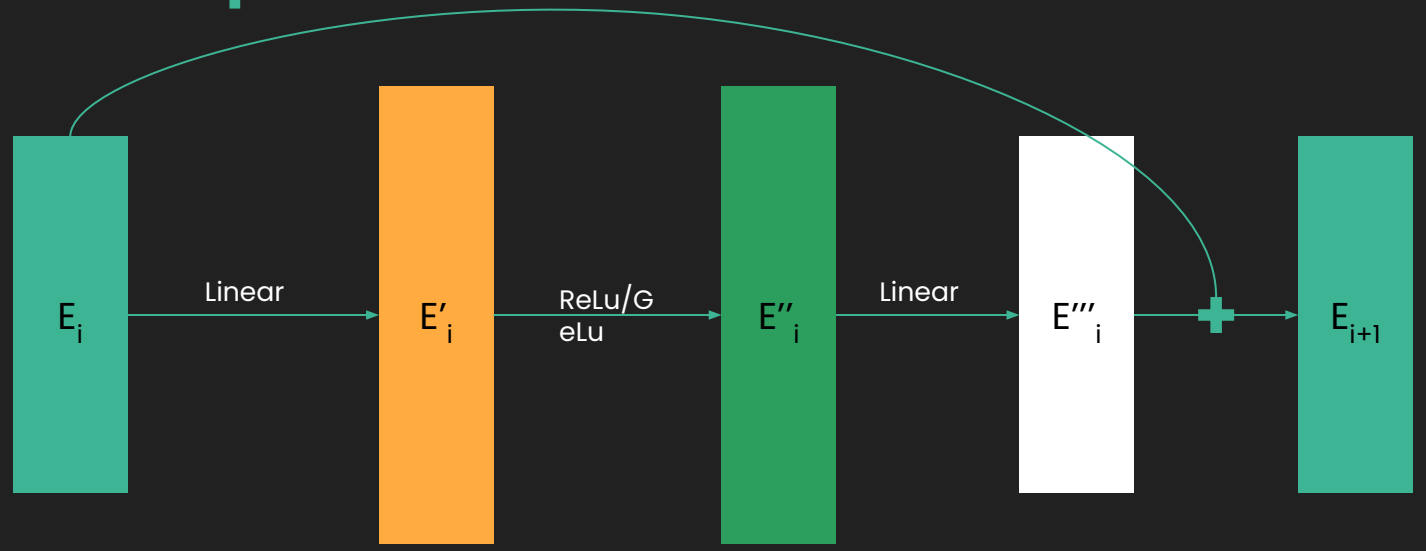
representative

1.1 + 0.0 +(-1).3 + 1.4 + 0.5 + (-1).6 + 7.1 + (-1).9 = -6

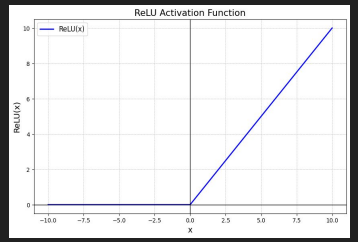
A single representative for each sub-window through convolution.

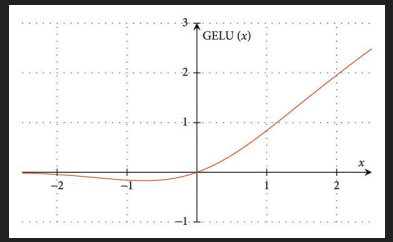
Helps in extracting global features in a computationally efficient method

Multi-Layer Perceptron



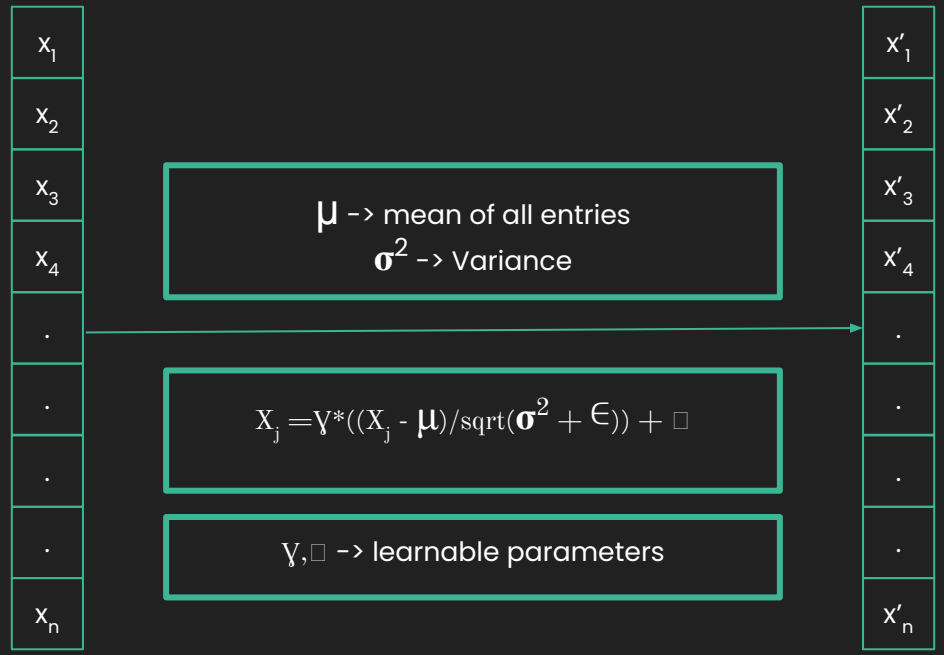
Actually asks the relevant questions



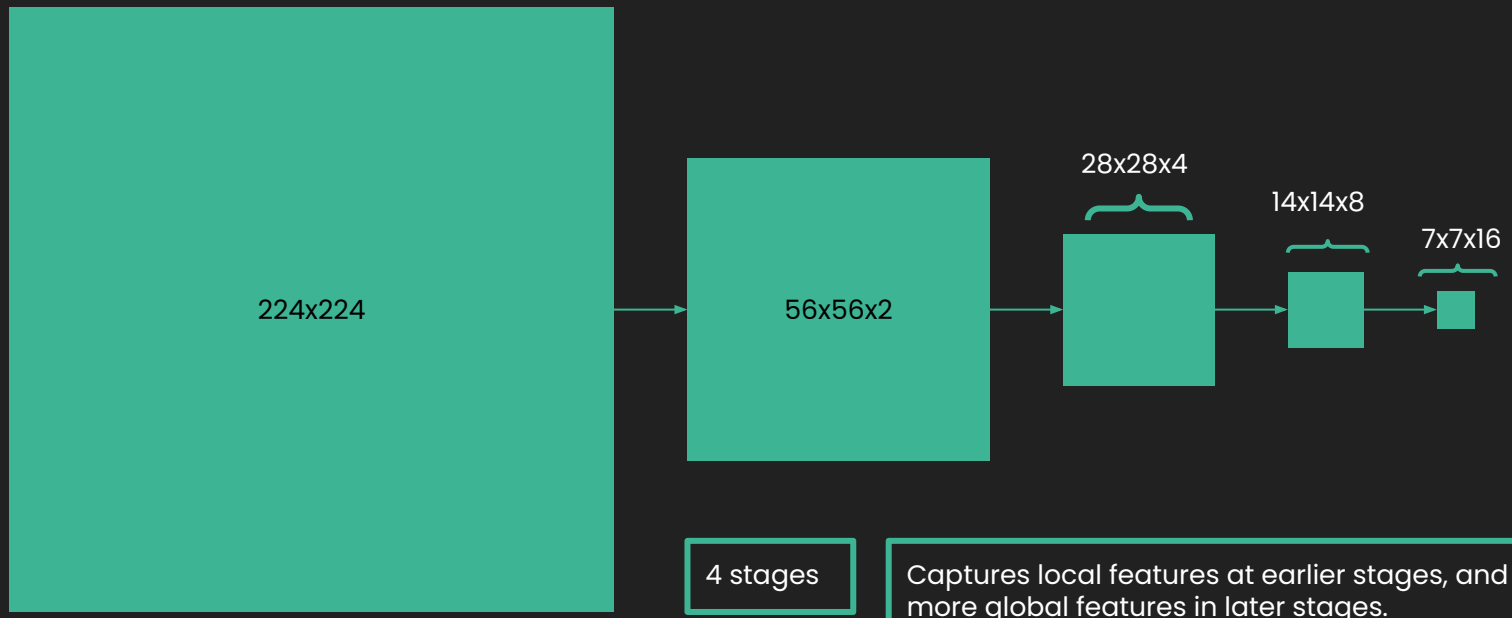


Layer Norm

Data is Normalized



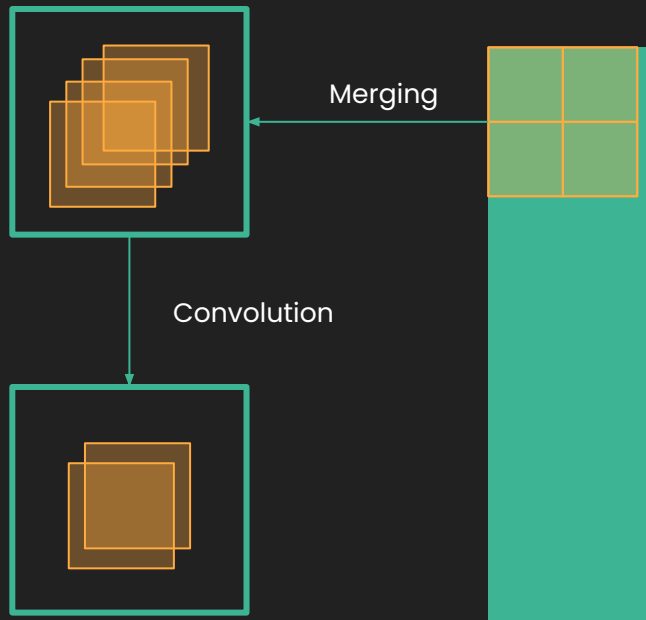
Reduction of Spatial Resolution - 1



4 stages

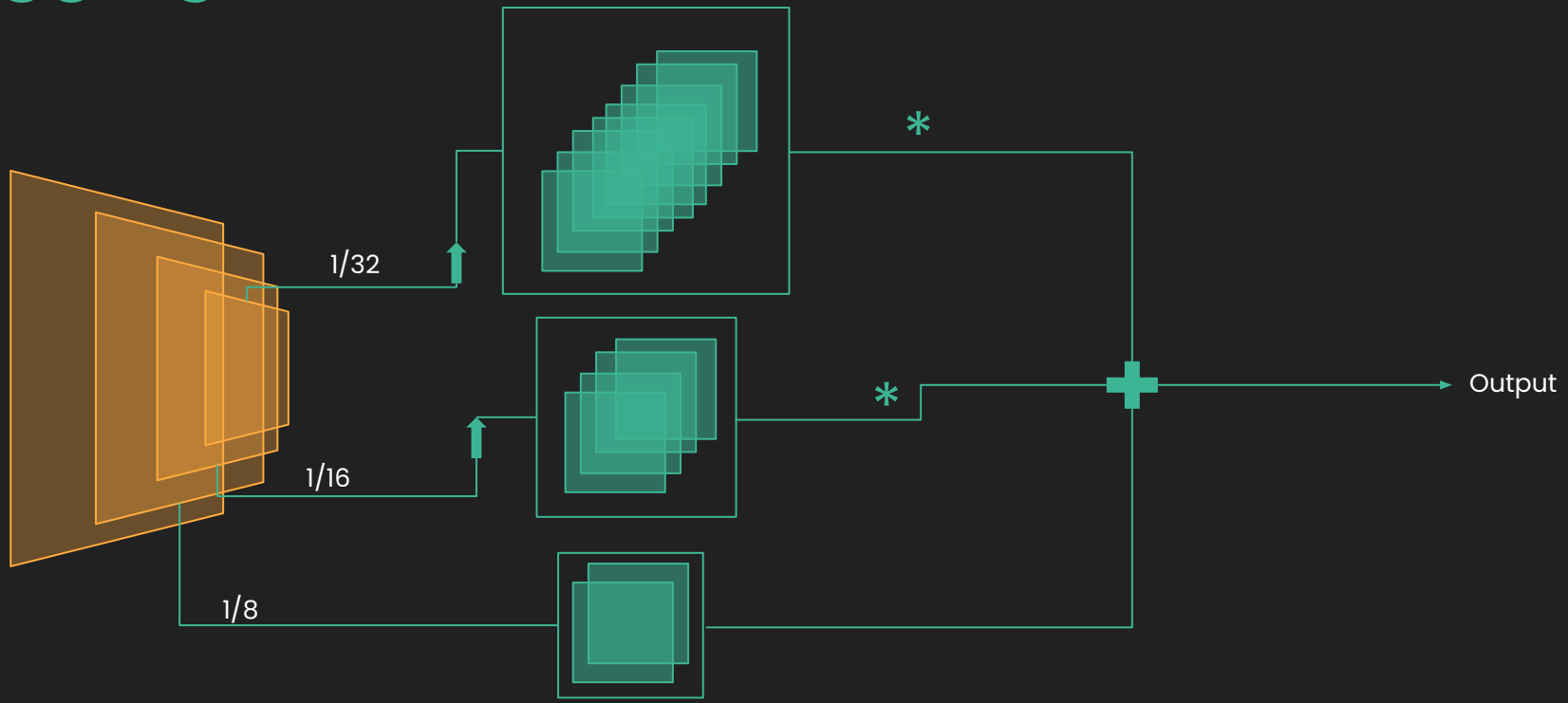
Captures local features at earlier stages, and more global features in later stages.

Reduction of Spatial Resolution - 2

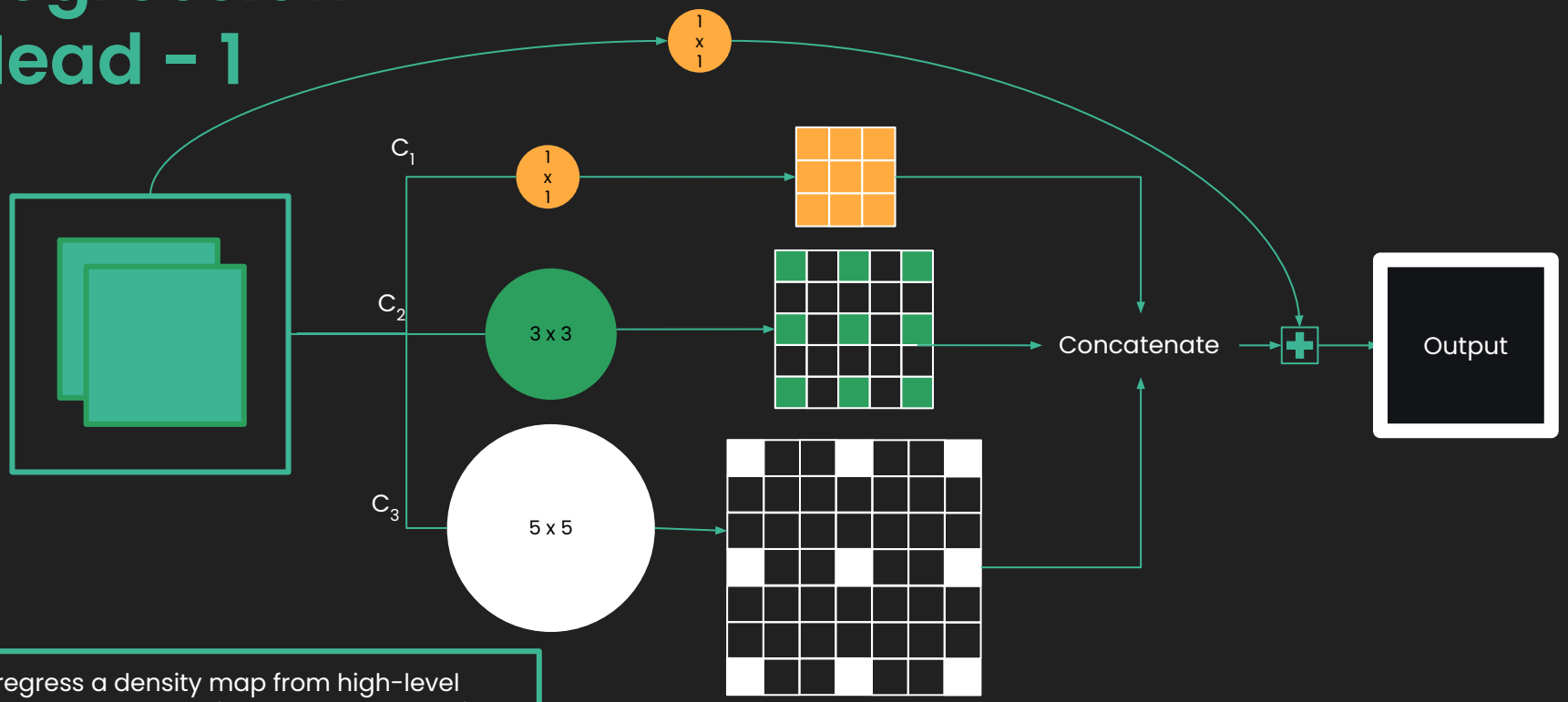


HxWx1

Pyramid Feature Aggregation

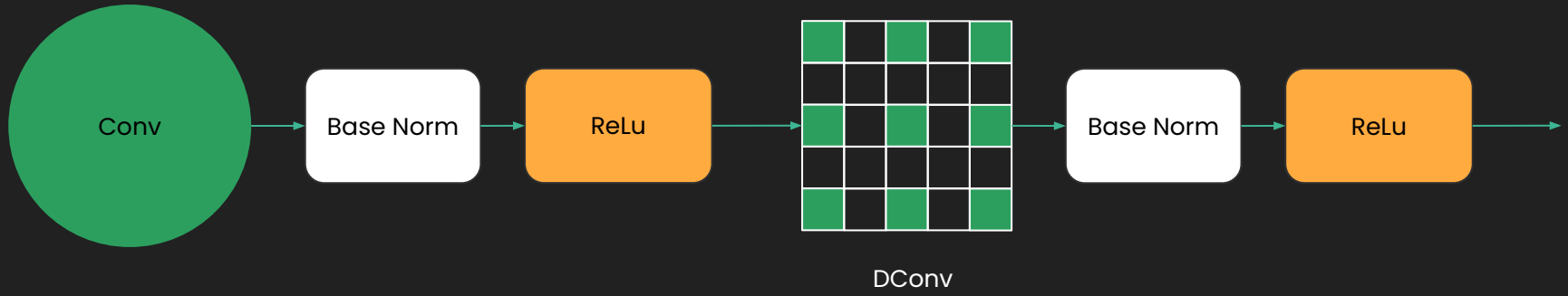


Regression Head - 1



To regress a density map from high-level feature maps, capturing small scale details while remaining computationally efficient.

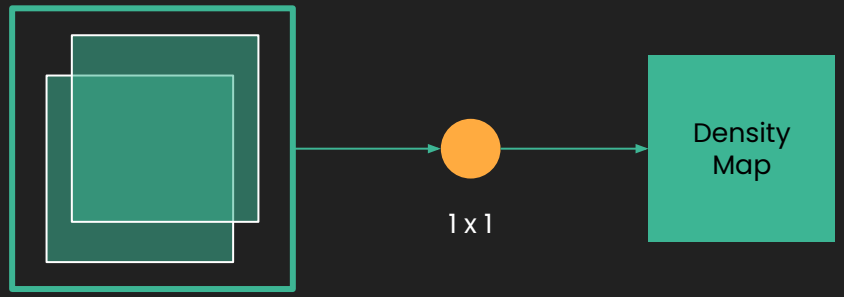
Regression Head - 2



3 parallel paths: C_1, C_2, C_3

Each captures details at different levels.

Achieves this by having different Receptive Fields



Loss function

$$\mathcal{L}_d = L_1(P, G) + \lambda_1 \mathcal{L}_{OT} + \lambda_2 L_2(D, D')$$

λ_1 = 0.01

λ_2 -> fine tuned

D -> predicted density map

D' -> ground truth density map

L_1 : Mean Absolute error -> |P-G|

L_2 : Mean squared error -> $\Sigma\ (D_{ij} - D_{ij}')^2$

λ_1 = 0.01

λ_2 -> fine tuned

L_{OT} =

Performance

Dataset	MAE	MSE
UCF_CC_50	168.7	234.5
ST_Part A	52.3	84.9
ST_Part B	6.2	9.9
UCF QNRF	82.8	142.3
NWPU-Crowd	79.8	344.4

Pre-trained with Imagenet 1k dataset

Outperforms State-of-the-art methods in all datasets (BCCT outperforms by slight margin in metric of MSE in ST_Part A)

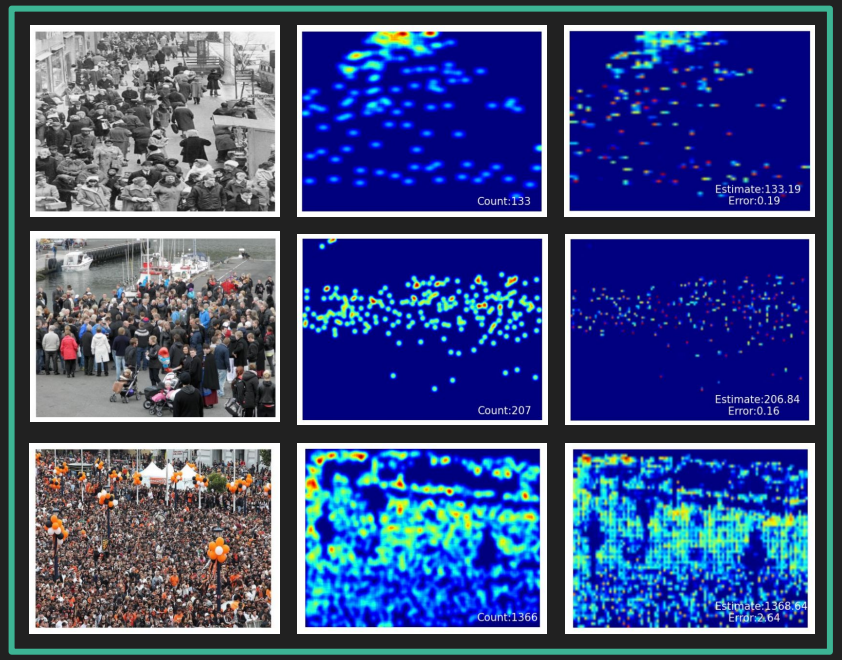
UCF_CC_50: Gray images with serious perspective distortions: still performs better with a margin

ST_Part A & B: intra-scene density variance very high: still performs well

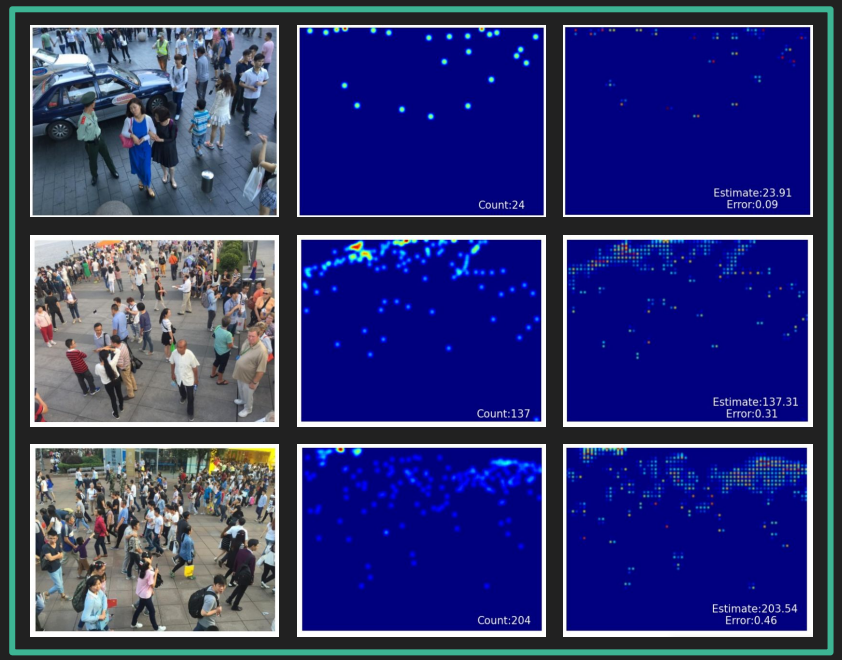
UCF QNRF: Large diversity in scenes and image sizes

NWPU-Crowd: Inter-scene scale and density variance

Visualisation Results



ShanghaiTech Part A



ShanghaiTech Part B